Automated Irregular Shaped Object Stacking via Monte Carlo Tree Search and Optimization

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***Abstract*—This paper presents a novel approach for automated stacking of irregularly shaped objects. The proposed method combines Monte Carlo Tree Search (MCTS) with optimization techniques to achieve stable and efficient stacking configurations.**

I. Introduction

Automating the process of stacking objects has long been studied in the context of packing and warehousing systems. Our contributions are as follows:

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The rest of the paper is organized as follows. Sec. III Sec. IV Sec. VI Sec. VII Sec. VIII

II. Preliminaries

A. Differentiable Support Function

Throughout the paper, we use the union of differentiable support function (DSF) to represent the 3D shapes of the target shape and the objects in the inventory. Given a compact convex body $C \subset \mathbb{R}^3$, the support function $h_C : \mathbb{R}^3 \backslash \{0\} \to \mathbb{R}$ is defined as

$$h_C(x) = \max_{s \in C} x \cdot s$$

where $\cdot$ denotes the standard inner product in $\mathbb{R}^3$. The work of Lee et al. [6] presented a differentiable version of the support function h_C that represents a smooth shape C.

The function $h_C(x)$ can be easily extended through rigid-body transformation as follows:

$$h_{\bar{C}(q)}(x) = h_C(R^\top x) + t \cdot x$$

for $q := (R, t) \in \text{SE}(3)$ where $\bar{C}(q) = \{Rp + t \mid p \in C\}$ denotes a transformed C under rigid-body transformation. Given two DSF represented convex bodies $C_1(q_1)$ and $C_2(q_2)$ with SE(3) poses q_1, q_2, the signed distance $\text{sd}(C_1(q_1), C_2(q_2))$ between the two convex bodies can be easily computed using Riemannian optimization techniques associated with $h_{C_1(q_1)}$ and $h_{C_2(q_2)}$ as presented in An, et al. [2]. Moreover, due to the smoothness of DSFs, the signed distance $\text{sd}(C_1(q_1), C_2(q_2))$ and the other contact information (e.g., normal vectors and contact points) are differentiable with respect to q_1 and q_2, which allows us to use gradient-based optimization techniques to optimize the placement poses of the objects in the later sections.

B. Envelope Theorem

Leveraged in many contact associated configuration optimization related works [7], [4], [8], the envelope theorem is a powerful tool that allows us to compute the gradient of an objective function with respect to a parameter when the objective function is defined as the optimal value of another optimization problem that depends on that parameter. Consider an optimization problem as follows:

$$\begin{aligned} \min_{q,\lambda} \quad & f(q, \lambda) \\ \text{s.t.} \quad & g_i(q, \lambda) \leq 0, \ i = 1, \cdots, m \\ & h_j(q, \lambda) = 0, \ j = 1, \cdots, n \end{aligned}$$

and the corresponding Lagrangian formulation:

$$L(q, \lambda, \nu, \zeta) = f(q, \lambda) + \sum_i \nu_i g_i(q, \lambda) + \sum_j \zeta_j h_j(q, \lambda)$$

where $\nu = (\nu_1, \cdots, \nu_m)$ and $\zeta = (\zeta_1, \cdots, \zeta_n)$ are dual variables with $\nu_i \geq 0$ for all $i = 1, \cdots, m$. For a fixed $\bar{q}$, let $\lambda^*(\bar{q})$, $\nu^*(\bar{q})$, $\zeta^*(\bar{q})$ be the optimal solution of the above optimization problem. Then *Envelop theorem* [3] states that the derivative of $\bar{f}(q) := f(q, \lambda^*(q))$ can be computed as follows:

$$\frac{d\bar{f}(q)}{dq} = \frac{\partial L(q, \lambda^*, \nu^*, \zeta^*)}{\partial q}. \tag{1}$$

C. Monte Carlo Tree Search (MCTS)

Given a state space $\mathcal{S}$, an action space $\mathcal{A}$, a transition function $p : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$, and a reward function $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, the Monte Carlo Tree Search (MCTS) is a popular algorithm for solving sequential decision-making problems that can be modeled as Markov Decision Processes (MDPs).

III. Problem Formulation and Framework Overview

A. Stacking as an Optimization Problem

Suppose that we are given a target shape $T \subset \mathbb{R}^3$ and n irregular shaped objects $O_1(q_1), O_2(q_2), \cdots, O_n(q_n) \subset \mathbb{R}^3$ where each $q_i \in \text{SE}(3)$ represents the pose of each object O_i in the world frame. The goal is to find optimal placement poses $q_1, \cdots, q_n$ for a set of irregularly shaped objects such that the final configuration approximates the target shape T while satisfying the stability and non-collision constraints. This problem can be formulated as an optimization problem:

$$\begin{aligned} \min_{q_1, \cdots, q_n \in \text{SE}(3)} \ & \text{Vol}\left(T \backslash \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \\ & c_{\text{phys}}(O_1(q_1), \cdots, O_n(q_n)) = 0 \end{aligned} \tag{2}$$

where $\text{Vol}(\cdot)$ is the volume of a 3D shape and $c_{\text{phys}}(\cdot)$ is a function that evaluates the physical feasibility of the configuration, including stability and non-collision constraints.

B. Markov Decision Process Formulation

Nevertheless, solving the optimization problem in Eq. (2) is extremely challenging due to the high-dimensional search space as the number of the objects n increases. In addition, the physical constraints involve complex contact dynamics which makes the problem even more difficult to solve.

To address the problem Eq. (2) in a tractable manner, we model the problem as a sequential decision-making process where we place one object at a time. Suppose that we have already placed k objects in the scene and denote the set of indices of the placed objects as $\mathcal{P}_k \subset \{1, \cdots, n\}$. Then, the k-th scene $\mathcal{X}_k := \{(i, q_i) \mid i \in \mathcal{P}_k\}$ is the collection of object indices and their corresponding placement poses that have been placed in the scene. Also, let the k-th inventory $\mathcal{I}_k := \{1, \cdots, n\} \backslash \mathcal{P}_k$ be the set of remaining object indices that have not been placed yet. Given $(\mathcal{X}_k, \mathcal{I}_k, T)$, we now need to select an index $i_{k+1} \in \mathcal{I}_k$ and find the corresponding placement pose q_{k+1} for the object such that the new scene $\mathcal{X}_{k+1} := \mathcal{X}_k \cup \{(i_{k+1}, q_{k+1})\}$ is physically feasible and approximates the target shape T.

The sequential decision-making process can be further modeled as a Markov Decision Process (MDP). We define the state space $\mathcal{S}$ as the set of states $s_k := (\mathcal{X}_k, \mathcal{I}_k, T)$, the action space $\mathcal{A}$ as the set of actions $a_k := (i_{k+1}, q_{k+1})$ representing the object index i_{k+1} and its placement pose q_{k+1}, the transition function $p : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ that maps a state-action pair to the next state, and the reward function $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ that evaluates the improvement in approximating the target shape T while satisfying the physical constraints. We will describe

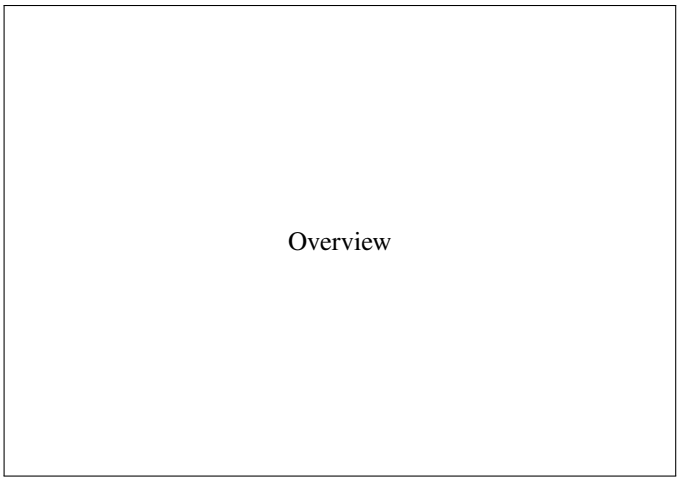


Fig. 1: Overview.

the details of the reward function design in the later section V. Therefore, the MDP can be defined as a tuple:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \gamma)$$

where $\gamma \in [0, 1]$ is the discount factor that determines the importance of future rewards.

C. Overall Structure of the Sequence Planner

The overall structure of the sequence planner is illustrated in Fig. 1. The sequence planner basically use Monte Carlo Tree Search (MCTS) to explore the space of possible stacking sequences, and during the search, it utilizes several components to sample the valid poses and evaluate the feasibility and quality of each candidate stacking sequence. The main components of the sequence planner include initial pose sampling module `InitialPoseSampling`, single object pose optimization module `PoseOptimization`, and structural simulation module `PhysicalSimulation`. We will further describe each component in detail in the later sections IV and V.

IV. Pose Optimization and Structural Simulation

In this section, we present the pose optimization and structural simulation components that are responsible for refining the candidate poses generated by the initial pose sampling component to find a physically feasible configuration that closely resembles the target shape.

A. Single Object Pose Optimization

Given a candidate pose $\hat{q}_{i_{k+1}}$ for an object $O_{i_{k+1}}$ in the inventory, the pose optimization component refines the candidate pose to find a locally optimal placement pose $q_{i_{k+1}}$ that likely leads to a physically feasible configuration and better approximates the target shape T. For a given scene $\mathcal{X}_k$, a target shape T, and a candidate object $O_{i_{k+1}}$, each pose refinement can be formulated as solving a constrained optimization problem:

$$\min_{\substack{q \in \mathrm{SE}(3), \\ \lambda \in \mathbb{R}^{3n_c}}} f_{\text{config}}(q \mid \mathcal{X}_k, T, O_{i_{k+1}}) + f_{\text{force}}(q, \lambda \mid \mathcal{X}_k, O_{i_{k+1}})$$
$$\text{s.t. } g_{\text{cone},j}(q, \lambda_j \mid \mathcal{X}_k, O_{i_{k+1}}) \leq 0, \quad j = 1, \cdots, n_c$$

where $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ is the vector of contact forces at the n_c contact points between the object and the scene. Each λ_j is represented in a Each $g_{\text{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\text{cone},j}(q, \lambda_j \mid \mathcal{X}_k, O_{i_{k+1}}) = \sqrt{\|J^t_{c,j}\lambda_j\|^2 + \epsilon} - \mu_j J^n_{c,j}\lambda_j$$

where $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J^t_{c,j}$ and $J^n_{c,j}$ are the tangential and normal components of the Jacobian of the contact constraint at the j-th contact point. The configuration dependent objective function $f_{\text{config}}(q \mid \mathcal{X}_k, T, O_{i_{k+1}})$ is a function that evaluates how well the configuration approximates the target shape T while satisfying the physical constraints such as collision avoidance of $O_{i_{k+1}}$ in the scene $\mathcal{X}_k$. The force dependent objective function $f_{\text{force}}(q, \lambda \mid \mathcal{X}_k, O_{i_{k+1}})$ is a function that evaluates whether the contact forces λ are physically feasible and contribute to a stable configuration. For simplicity, we will omit the dependency of f_{config}, f_{force} and $g_{\text{cone},j}$ on $\mathcal{X}_k$, T, and $O_{i_{k+1}}$ in the following subsections.

Given a fixed pose q, the contact forces λ can be efficiently optimized by solving a quadratic programming problem under the second-order cone constraints. We will describe the details of the force optimization in the next subsection. Denoting $\lambda^*(q)$ as the optimal contact forces for a given pose q, we can define the force dependent objective function as $f_{\text{force}}(q) := f_{\text{force}}(q, \lambda^*(q))$. Moreover, denote $\nu^*(q) \in \mathbb{R}^{n_c}$ as the optimal dual variables corresponding to the friction cone constraints. Then, using the *Envelope theorem* (1), we can compute the gradient of $f_{\text{force}}(q)$ with respect to q as follows:

$$\frac{df_{\text{force}}(q)}{dq} = \frac{\partial f_{\text{force}}(q, \lambda^*(q))}{\partial q} + \sum_{j=1}^{n_c} \nu^*_j(q) \frac{\partial g_{\text{cone},j}(q, \lambda^*_j(q))}{\partial q}.$$

Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with Riemannian BFGS Hessian approximation [5] to find a locally optimal placement pose $q_{i_{k+1}}$ that minimizes the objective function $f_{\text{config}}(q) + f_{\text{force}}(q)$ while satisfying the friction cone constraints $g_{\text{cone},j}(q, \lambda^*_j(q)) \leq 0$ for $j = 1, \cdots, n_c$.

B. Force Optimization

For indices $l, m \in \mathcal{P}_k$, let $I_{l,m} \subset \{1, \cdots, n_c\}$ be the set of contact point indices associated with the objects O_l, O_m, and let $I_l := \bigcup_{m \in \mathcal{P}_k} I_{l,m}$ be the set of contact point indices associated with the object O_l. Then, the force dependent

objective $f_{\text{force}}(q, \lambda)$ is defined as the following:

$$f_{\text{force}}(q,\lambda) = \sum_{l \in \mathcal{P}_k} \|J_{c,I_l}^{\top}(q)\lambda_{I_l} + \lambda_{\text{ext},l}\|^2 + \rho \sum_{j=1}^{n_c} \|\lambda_j\|^2 + \sum_{l \in \mathcal{P}_k} \sum_{j \in I_{l,i_{k+1}}} \text{sd}_j(q)^2 \|\lambda_j\|^2 \tag{3}$$

where $J_{c,I_l}(q) \in \mathbb{R}^{3|I_l| \times 6}$ is the Jacobian of the contact constraints associated with the object O_l at the contact points in I_l, $\lambda_{I_l} \in \mathbb{R}^{3|I_l|}$ is the vector of contact forces at the contact points in I_l, and $\lambda_{\text{ext},l} \in \mathbb{R}^6$ is the external wrench acting on the object O_l. The first term in $f_{\text{force}}(q, \lambda)$ penalizes the violation of the static equilibrium constraints for each object in the scene. The second term with $\rho > 0$ is a regularization term to avoid unnecessarily large contact forces which is not physically meaningful. The third term penalizes the complementarity conditions on the contact forces between the candidate object $O_{i_{k+1}}$ and the objects in the scene $\mathcal{P}_k$ based on the signed distance $\text{sd}_j(q)$ between the two objects at the j-th contact point which can be computed as described in Section II. Note that $\text{sd}_j(q)$ is differentiable which allows us to apply *Envelope theorem* to compute the gradient of $f_{\text{force}}(q)$ with respect to q as described in the previous subsection.

C. Structural Simulation

V. Sequence Planning

A. Initial Pose Sampling

The initial pose sampler generates a set of candidates poses and the corresponding object indices in the inventory based on the target shape and the current scene configuration. Initially, using the heightmap $H_T : \mathbb{R}^2 \to \mathbb{R}$ of the target shape, the heightmap $H_{\mathcal{X}} : \mathbb{R}^2 \to \mathbb{R}$ of the current scene configuration and the the sampler construct a planar score map that indicates the potential placement locations for each object in the inventory.

VI. Evaluations

VII. Experiments

VIII. Conclusion

Acknowledgments

References

[1] P. A. Absil, C. G. Baker, and K. A. Gallivan. Trust-region methods on riemannian manifolds. *Foundations of Computational Mathematics*, 7:303–330, 2007.

[2] S. An, S. Lee, J. Lee, S. Park, and D. Lee. Collision detection between smooth convex bodies via riemannian optimization framework. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 12464–12471. IEEE, 2024.

[3] M. Carter. *Foundations of mathematical economics*. MIT press, 2001.

[4] J. Chen, Y. Ke, and H. Wang. BODex: Scalable and efficient robotic dexterous grasp synthesis using bilevel optimization. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1–8. IEEE, 2025. doi: 10.1109/ICRA55743.2025.11127930.

[5] W. Huang, P.-A. Absil, and K. A. Gallivan. A riemannian BFGS method without differentiated retraction for nonconvex optimization problems. *SIAM Journal on Optimization*, 28(1):470–495, 2018. doi: 10.1137/17M1127582.

[6] J. Lee, M. Lee, and D. Lee. Uncertain pose estimation during contact tasks using differentiable contact features. *RSS 2023-Robotics: Science and Systems*, 2023.

[7] J. Lee, S. Park, M. Lee, and D. Lee. Efficient gradient-based inference for manipulation planning in contact factor graphs. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 12464–12471. IEEE, 2025.

[8] R. Zurbrügg, A. Cramariuc, and M. Hutter. GraspQP: Differentiable optimization of force closure for diverse and robust dexterous grasping. In *Proceedings of the 9th Conference on Robot Learning (CoRL)*, volume 305 of *Proceedings of Machine Learning Research*, pages 2583–2602. PMLR, 2025.